THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solid metallic cone of base radius 12 cm and height 6 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

(a) 6 cm **(b)** 7 cm **(c)** 12 cm **(d)** 5 cm

2. A toy is a cone of base radius 14 cm and height 48 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 3344 cm² **(b)** 3432 cm² **(c)** 4048 cm² **(d)** 2200 cm²

3. A capsule is a cylinder of radius 14 cm and length 28 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 9856 cm² **(b)** 3696 cm² **(c)** 4928 cm² **(d)** 6160 cm²

4. A solid is a cube of edge 21 cm surmounted by a hemisphere of diameter 21 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 3339 cm² **(b)** 2299.5 cm² **(c)** 2646 cm² **(d)** 2992.5 cm²

5. From a solid cylinder of height 72 cm and base radius 21 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

(a) 17226 cm² **(b)** 12276 cm² **(c)** 15840 cm² **(d)** 14454 cm²

6. A wooden article is made by scooping out a hemisphere of radius 10.5 cm from each end of a solid cylinder of height 12 cm and base radius 10.5 cm. The total surface area of the article is: 1

(a) 2178 cm² **(b)** 792 cm² **(c)** 1485 cm² **(d)** 2871 cm²

7. A solid metallic cylinder of base radius 9 cm and height 24 cm is melted and recast into solid spheres of radius 3 cm. The number of spheres obtained is: 1

(a) 108 **(b)** 52 **(c)** 60 **(d)** 54

8. A solid metallic cone of base radius 4 cm and height 54 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

(a) 5 cm **(b)** 6 cm **(c)** 12 cm **(d)** 7 cm

9. A toy is a cone of base radius 28 cm and height 45 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 9592 cm² **(b)** 4664 cm² **(c)** 8888 cm² **(d)** 12056 cm²

10. A capsule is a cylinder of radius 10.5 cm and length 12 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 2871 cm² **(b)** 4356 cm² **(c)** 1485 cm² **(d)** 2178 cm²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 6 cm, and no water spills. Find the rise in the water level. 2

12. A solid metallic sphere of radius 9 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. Find the length of the wire. 2

13. A solid metal cuboid 24 cm by 24 cm by 9 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 2

14. A solid sphere of radius 9 cm is completely dropped into water in a cylindrical vessel of base radius 9 cm, and no water spills. Find the rise in the water level. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A toy is a cone of base radius 21 cm and height 35 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

16. A capsule is a cylinder of radius 10.5 cm and length 15 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

17. From a solid cylinder of height 6 cm and base radius 14 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

OR

A toy is a cone of base radius 10.5 cm and height 21 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

18. A capsule is a cylinder of radius 21 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

OR

From a solid cylinder of height 12 cm and base radius 21 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Water from a full cylindrical tank of radius 14 cm and height 6 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

OR

A conical vessel of radius 6 cm and height 24 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

20. Water from a full cylindrical tank of radius 7 cm and height 28 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

OR

A conical vessel of radius 6 cm and height 30 cm is full of water. The water is poured into a cylindrical vessel of base radius 6 cm. Find the height of the water in the cylindrical vessel. 5

END OF QUESTION PAPER — Total: 40 Marks